WORKED SOLUTION

STEP III 2024, Question 2

Pure •

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Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

The key idea

Every expression under a square root here is a quadratic that is always positive, and both sides of each inequality are non-negative, so we may square without changing the solution set. Squaring turns each problem into a statement about the sign of a quadratic (or, when the x^2 terms cancel, a linear function). Geometrically, $y = \sqrt{4x^2 - 8x + 64}$ is the top half of a hyperbola, and the lines in the question are chords and asymptotes of it.

Solution

Part (i)

Completing the square, $4x^2 - 8x + 64 = 4(x-1)^2 + 60 > 0$, so the square root is defined for every x. Both sides of each inequality are non-negative, and for $A, B \geq 0$ we have $A \leq B \iff A^2 \leq B^2$.

(a) Squaring,

$$\begin{aligned} 4x^2 - 8x + 64 \leq x^2 + 16x + 64 &\iff 3x^2 - 24x \leq 0 \\ &\iff 3x(x-8) \leq 0 \iff 0 \leq x \leq 8. \end{aligned}$$

(b) Squaring,

$$\begin{aligned} 4x^2 - 8x + 64 \leq 9x^2 - 48x + 64 &\iff 0 \leq 5x^2 - 40x \\ &\iff 5x(x-8) \geq 0 \iff x \leq 0 \text{ or } x \geq 8. \end{aligned}$$

In both parts, equality holds at $x = 0$ (all sides equal 8) and at $x = 8$ (all sides equal 16).

Part (ii)

(a) Using the difference of two squares,

$$\left(\sqrt{4x^2 - 8x + 64} + 2(x-1)\right)\mathrm{f}(x) = (4x^2 - 8x + 64) - 4(x-1)^2 = 60.$$

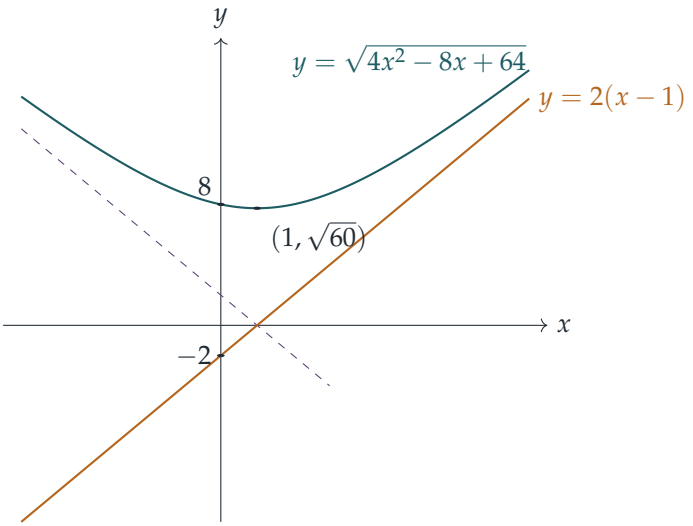
For $x > 1$ the bracket is positive, so we may divide:

$$f(x) = \frac{60}{\sqrt{4x^2 - 8x + 64} + 2(x-1)}.$$

As $x \to \infty$ the denominator is at least $2(x-1)$, which tends to infinity, so $f(x) \to 0$, as required.

(b) Write $y = \sqrt{4(x-1)^2 + 60}$. Then $y > 0$ and $y^2 - 4(x-1)^2 = 60$, so the curve is the upper branch of a hyperbola centred at $(1, 0)$. It is symmetric about $x = 1$, with minimum value $\sqrt{60} = 2\sqrt{15} \approx 7.75$ at $x = 1$, and it crosses the y-axis at $(0, 8)$.

Since $y > \sqrt{4(x-1)^2} = 2|x-1|$, the curve lies strictly above both lines $y = \pm 2(x-1)$. Part (a) says it approaches $y = 2(x-1)$ as $x \to \infty$. By the symmetry about $x = 1$, it approaches $y = -2(x-1)$ as $x \to -\infty$. The line $y = 2(x-1)$ has gradient 2 and passes through $(1, 0)$ and $(0, -2)$.



The dashed line is the other asymptote, $y = -2(x-1)$.

Part (iii)

Again $4x^2 - 5x + 4 = 4(x - \frac{5}{8})^2 + \frac{39}{16} > 0$, so the square root is always defined. Squaring both (non-negative) sides, the inequality is equivalent to

$$(m^2 - 4)x^2 + (2mc + 5)x + (c^2 - 4) \geq 0. \qquad (*)$$

Ask yourself What kind of expression on the left of $(*)$ could be non-negative exactly when $x \geq 3$?

If $m^2 - 4 > 0$, the left-hand side tends to $+\infty$ as $x \to -\infty$, so very negative x would satisfy $(*)$. If $m^2 - 4 < 0$, it tends to $-\infty$ as $x \to +\infty$, so large x would fail. Neither is

allowed, so $m^2 = 4$, and $(*)$ becomes linear. A linear inequality $ax + b \geq 0$ has solution set $\{x : x \geq 3\}$ exactly when $a > 0$ and $3a + b = 0$.

Take $m = 2$. Then $a = 4c + 5$ and $b = c^2 - 4$, so we need

$$3(4c + 5) + c^2 - 4 = c^2 + 12c + 11 = (c + 1)(c + 11) = 0, \qquad 4c + 5 > 0.$$

The root $c = -11$ gives $a = -39 < 0$, which is no good. The root $c = -1$ gives $a = 1 > 0$. So

$$m = 2, \qquad c = -1$$

works. To confirm, $(2x - 1)^2 - (4x^2 - 5x + 4) = x - 3$, which is non-negative exactly when $x \geq 3$. (Taking $m = -2$, $c = 1$ gives the same $|mx + c|$ and is equally valid.)

Part (iv)

Write $g(x) = x^2 + px + q$ and $L(x) = mx + c$. Since $|g| \leq L \iff -L \leq g \leq L$, the inequality holds exactly when both

$$g(x) - L(x) \leq 0 \qquad \text{and} \qquad g(x) + L(x) \geq 0.$$

Both $g - L$ and $g + L$ are quadratics with leading coefficient 1.

- The set where $g - L \leq 0$ is a closed interval $[a, b]$ (possibly empty or a single point).
- The set where $g + L \geq 0$ is everything except an open interval (r, s). If $g + L$ has no real roots, nothing is removed.

So the solution set is $[a, b]$ with (r, s) removed. For this to be $[-5, 1] \cup [5, 7]$ we need $a = -5$ and $b = 7$ (the outer ends), and the removed gap must be exactly $(1, 5)$, so $r = 1$ and $s = 5$. Hence

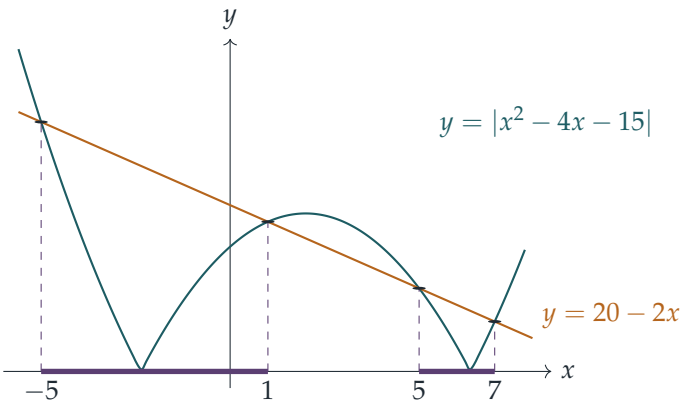
$$g(x) - L(x) = (x + 5)(x - 7) = x^2 - 2x - 35,$$
$$g(x) + L(x) = (x - 1)(x - 5) = x^2 - 6x + 5.$$

Adding and halving gives $g(x) = x^2 - 4x - 15$. Subtracting and halving gives $L(x) = -2x + 20$. Therefore

$$p = -4, \qquad q = -15, \qquad m = -2, \qquad c = 20.$$

Every step above is reversible, so with these values the solution set really is $[-5, 1] \cup [5, 7]$: we need $(x + 5)(x - 7) \leq 0$ and $(x - 1)(x - 5) \geq 0$. These values are also the only ones that work.

In the picture, g has roots $2 \pm \sqrt{19} \approx -2.36,\ 6.36$. The line $y = 20 - 2x$ meets the outer arms of $|g|$ at $x = -5$ and $x = 7$, and cuts through the reflected hump at $x = 1$ and $x = 5$.



Checks

- In (ii)(a), at $x = 100$ we get $f(100) = \sqrt{39264} - 198 \approx 0.151$, matching $60/(\sqrt{39264} + 198) \approx 0.151$.
- In (iv), at $x = 0$: $|-15| = 15 \leq 20$, so 0 is a solution. At $x = 3$: $|9 - 12 - 15| = 18 > 14$, so 3 is not. At $x = 8$: $|64 - 32 - 15| = 17 > 4$, so 8 is not. These match $[-5, 1] \cup [5, 7]$. A computer check over a fine grid of x confirmed all four solution sets.

Common slips

- Squaring without saying why it is allowed. It is valid here only because both sides are non-negative.
- In (iii), keeping a genuine quadratic. A quadratic inequality can never have a one-sided half-line as its solution set, so the x^2 terms must cancel.
- In (iv), pairing the endpoints wrongly: the outer ends $-5, 7$ belong to $g - L$, and the inner ends $1, 5$ (the gap) to $g + L$.

Beyond STEP

The curve $y^2 - 4(x-1)^2 = 60$ is a hyperbola, and part (ii) shows how an asymptote is found: rationalise the difference between the curve and the candidate line. The same trick, multiplying by a "conjugate", is a standard tool for limits in first-year analysis. Writing $|g| \leq L$ as two one-sided conditions, as in (iv), is how inequalities with absolute values are routinely handled there. Abbott's *Understanding Analysis* is a good next step.

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